The Iwahori–Hecke Algebra from Lines in Projective Space

Notes

Abstract

We derive the Iwahori–Hecke algebra $H_n(q)$ from the double coset algebra $\mathcal{H}(\mathrm{GL}_n(\mathbb{F}_q), B_n(\mathbb{F}_q))$ by computing how the double coset operators act on the flag variety. The quadratic relation $(T_s - q)(T_s + 1) = 0$ is derived by counting incidences of lines in $\mathbb{P}^1(\mathbb{F}_q)$. The braid relation is derived from the geometry of flags in $\mathbb{F}_q^3$.

Contents

1 Setup **1**
- 1.1 The groups . . . 1
- 1.2 The flag variety . . . 1
- 1.3 The Weyl group . . . 2
- 1.4 The Bruhat decomposition . . . 2

2 The quadratic relation from $\mathbb{P}^1(\mathbb{F}_q)$ **2**
- 2.1 The action on $\mathbb{P}^1(\mathbb{F}_q)$ . . . 2
- 2.2 Deriving the quadratic relation . . . 3
- 2.3 The limit $q \to 1$ . . . 3

3 The braid relation from flags in $\mathbb{F}_q^3$ **4**
- 3.1 The flag variety for GL_3 . . . 4
- 3.2 Counting . . . 4
- 3.3 The braid relation . . . 4

4 The isomorphism **5**
- 4.1 The Iwahori–Hecke algebra by generators and relations . . . 5
- 4.2 The isomorphism theorem . . . 5

5 Summary **6**

1 Setup

1.1 The groups

Let q be a prime power and $G = \mathrm{GL}_n(\mathbb{F}_q)$ the group of invertible $n \times n$ matrices over $\mathbb{F}_q$. The *Borel subgroup* $B = B_n(\mathbb{F}_q)$ is the subgroup of upper triangular matrices:

$$B = \left\{ \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{pmatrix} : a_{ii} \in \mathbb{F}_q^*,\ a_{ij} \in \mathbb{F}_q \text{ for } i < j \right\}.$$

The order is $|B| = (q-1)^n \cdot q^{\binom{n}{2}}$ (n diagonal entries from $\mathbb{F}_q^*$ and $\binom{n}{2}$ strictly upper-triangular entries from $\mathbb{F}_q$).

1.2 The flag variety

B is the stabiliser of the *standard flag*

$$F_\bullet : \quad \{0\} \subset \langle e_1 \rangle \subset \langle e_1, e_2 \rangle \subset \cdots \subset \langle e_1, \ldots, e_n \rangle = \mathbb{F}_q^n,$$

where $e_1, \ldots, e_n$ is the standard basis. The coset space G/B is the *complete flag variety*: the set of all complete flags

$$\{0\} = V_0 \subset V_1 \subset V_2 \subset \cdots \subset V_n = \mathbb{F}_q^n, \qquad \dim V_i = i.$$

For $n = 2$: a complete flag is just a line $V_1 \subset \mathbb{F}_q^2$, so $G/B = \mathbb{P}^1(\mathbb{F}_q)$, the projective line with $q+1$ points.

For $n = 3$: a complete flag is a line V_1 inside a plane V_2 inside $\mathbb{F}_q^3$, so G/B is the set of incident line-plane pairs.

1.3 The Weyl group

The *Weyl group* of GL_n is $W = S_n$, realised as permutation matrices. The *simple reflections* are the adjacent transpositions $s_i = (i,\ i{+}1)$ for $i = 1, \ldots, n-1$, represented by the permutation matrix that swaps rows i and $i{+}1$.

1.4 The Bruhat decomposition

Theorem 1.1 (Bruhat)**.** $\mathrm{GL}_n(\mathbb{F}_q)$ *decomposes as a disjoint union of double cosets:*

$$\mathrm{GL}_n(\mathbb{F}_q) = \bigsqcup_{w \in S_n} BwB.$$

The double cosets are indexed by the Weyl group $W = S_n$.

Proof for $n = 2$. $G = \mathrm{GL}_2(\mathbb{F}_q)$, $B =$ upper triangular, $W = \{e, s\}$ where $s = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$.

Any $g = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in G$:

- If $c = 0$: then g is upper triangular, so $g \in B = BeB$.

- If $c \neq 0$: we can write $g = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) = \left(\begin{smallmatrix} ac^{-1} & * \\ 1 & * \end{smallmatrix}\right) \cdot \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right) \cdot \left(\begin{smallmatrix} c & d \\ 0 & * \end{smallmatrix}\right)$ (filling in the $*$'s to make the equation hold). So $g \in BsB$.

Thus $\mathrm{GL}_2 = B \sqcup BsB$: every matrix is either upper triangular or has nonzero $(2,1)$-entry, and in the latter case it belongs to BsB. □

It follows that $\mathcal{H}(\mathrm{GL}_n(\mathbb{F}_q), B)$ has dimension $|W| = n!$, with basis $\{T_w : w \in S_n\}$ where T_w is the characteristic function of BwB (suitably normalised).

2 The quadratic relation from $\mathbb{P}^1(\mathbb{F}_q)$

We work with $G = \mathrm{GL}_2(\mathbb{F}_q)$ and $B =$ upper triangular. The Hecke algebra $\mathcal{H}(G, B)$ is 2-dimensional with basis $\{T_e, T_s\}$. We derive the quadratic relation by computing how T_s acts on the projective line.

2.1 The action on $\mathbb{P}^1(\mathbb{F}_q)$

$\mathbb{P}^1(\mathbb{F}_q)$ has $q+1$ points (the lines through the origin in $\mathbb{F}_q^2$). Let δ_L denote the basis vector for the point $L \in \mathbb{P}^1(\mathbb{F}_q)$ in the permutation representation $k[\mathbb{P}^1(\mathbb{F}_q)] = k[G/B]$. The Hecke algebra acts on this space.

T_e is the identity: $T_e \cdot \delta_L = \delta_L$.

T_s acts by sending each point to the sum of all *other* points:

$$T_s \cdot \delta_L = \sum_{L' \neq L} \delta_{L'}. \tag{1}$$

Proof of (1). The operator T_s corresponds to the double coset BsB. It acts on δ_L by summing over all cosets gB with $g^{-1}L' = L$ and $g \in BsB$, for varying L'. Since B stabilises the standard line $\langle e_1 \rangle$ and s swaps it to $\langle e_2 \rangle$, the double coset BsB consists of all matrices that send $\langle e_1 \rangle$ to a line $\neq \langle e_1 \rangle$. By the transitivity of B on lines $\neq \langle e_1 \rangle$ (upper triangular matrices act transitively on the q affine points of $\mathbb{P}^1$), each line $L' \neq L$ appears exactly once. □

Equivalently: T_s is the adjacency matrix of the complete graph K_{q+1} on $\mathbb{P}^1(\mathbb{F}_q)$, minus the identity. In matrix form, $T_s = J - I$ where J is the all-ones matrix and I the identity.

2.2 Deriving the quadratic relation

Theorem 2.1. $T_s^2 = (q-1)\,T_s + q\,T_e$.

Proof. Apply T_s twice to δ_L for any point $L \in \mathbb{P}^1(\mathbb{F}_q)$.

First application: by (1), $T_s \cdot \delta_L = \sum_{L' \neq L} \delta_{L'}$ (a sum of q terms).

Second application: apply T_s to each $\delta_{L'}$:

$$T_s^2 \cdot \delta_L = \sum_{L' \neq L} T_s \cdot \delta_{L'} = \sum_{L' \neq L} \sum_{L'' \neq L'} \delta_{L''}.$$

Now count how many times each point L'' appears in this double sum:

- $L'' = L$: for each $L' \neq L$, the inner sum includes $L'' = L$ (since $L \neq L'$). There are q such L'. Contribution: $q \cdot \delta_L$.
- $L'' \neq L$: for a fixed $L'' \neq L$, the inner sum includes L'' whenever $L' \neq L''$. Among the q choices of L' (all $\neq L$), exactly one is excluded ($L' = L''$), leaving $q-1$ choices. Contribution: $(q-1) \cdot \delta_{L''}$ for each $L'' \neq L$.

Combining:

$$T_s^2 \cdot \delta_L = q \cdot \delta_L + (q-1) \sum_{L'' \neq L} \delta_{L''} = q \cdot T_e \cdot \delta_L + (q-1) \cdot T_s \cdot \delta_L.$$

Since this holds for every L, we have $T_s^2 = (q-1)\, T_s + q\, T_e$. □

Corollary 2.2. *The quadratic relation factors as*

$$(T_s - q)(T_s + 1) = 0. \tag{2}$$

Proof. $T_s^2 - (q-1)T_s - q = T_s^2 - qT_s + T_s - q = T_s(T_s - q) + (T_s - q) = (T_s + 1)(T_s - q) = 0$. □

Remark 2.3. The two roots $T_s = q$ and $T_s = -1$ are the eigenvalues of T_s on $k[\mathbb{P}^1(\mathbb{F}_q)]$:

- $T_s = q$ on the trivial representation (the constant function $\sum_L \delta_L$): T_s sends it to q times itself, since each point has q neighbours.
- $T_s = -1$ on the q-dimensional complement (functions summing to zero): $T_s \cdot f = (J - I) \cdot f = -f$ since $J \cdot f = 0$ for f orthogonal to the constant function.

2.3 The limit $q \to 1$

At $q = 1$ (formally): the relation becomes $(T_s - 1)(T_s + 1) = 0$, i.e., $T_s^2 = 1$. This is the involution relation for the symmetric group. The projective line $\mathbb{P}^1(\mathbb{F}_1)$ has $1 + 1 = 2$ points, and T_s swaps them—exactly the transposition $(1\ 2) \in S_2$.

3 The braid relation from flags in $\mathbb{F}_q^3$

For $G = \mathrm{GL}_3(\mathbb{F}_q)$, the Weyl group is $W = S_3$ with generators $s_1 = (1\ 2)$ and $s_2 = (2\ 3)$. The Hecke algebra $\mathcal{H}(G, B)$ has dimension 6 with basis $\{T_w : w \in S_3\}$.

We need to verify: $T_{s_1} T_{s_2} T_{s_1} = T_{s_2} T_{s_1} T_{s_2}$ (the braid relation).

3.1 The flag variety for GL_3

A complete flag in $\mathbb{F}_q^3$ is an incident pair (line ℓ, plane π) with $\ell \subset \pi \subset \mathbb{F}_q^3$. The standard flag is $F_\bullet = (\langle e_1 \rangle,\ \langle e_1, e_2 \rangle)$.

The operators T_{s_1} and T_{s_2} act on flags by modifying one component:

- T_{s_1} changes the *line* ℓ while keeping the plane π fixed: it sums over all lines $\ell' \neq \ell$ in π.
- T_{s_2} changes the *plane* π while keeping the line ℓ fixed: it sums over all planes $\pi' \neq \pi$ containing ℓ.

Why T_{s_1} changes only the line. The simple reflection s_1 swaps e_1 and e_2. So BsB consists of matrices that move $\langle e_1 \rangle$ to another line inside $\langle e_1, e_2 \rangle$, while preserving $\langle e_1, e_2 \rangle$ itself. That is, T_{s_1} replaces the line component of the flag by another line in the same plane. □

3.2 Counting

In $\mathbb{F}_q^3$:

- A plane $\pi \cong \mathbb{F}_q^2$ contains $q+1$ lines. So T_{s_1} sums over q lines (all lines in π except ℓ).
- Through a line ℓ, there are $q+1$ planes in $\mathbb{F}_q^3$. So T_{s_2} sums over q planes (all planes through ℓ except π).

Each operator is a "local T_s": T_{s_1} acts like the $\mathbb{P}^1(\mathbb{F}_q)$-operator from Section 2 on the lines within a fixed plane, and T_{s_2} acts like it on the planes through a fixed line. Each satisfies the quadratic relation $(T_{s_i} - q)(T_{s_i} + 1) = 0$ by the same counting argument as Theorem 2.1.

3.3 The braid relation

Theorem 3.1. $T_{s_1}T_{s_2}T_{s_1} = T_{s_2}T_{s_1}T_{s_2}$.

Proof. Apply both sides to a flag (ℓ, π) and show they give the same result.

Left side: $T_{s_1}T_{s_2}T_{s_1}$. Starting from (ℓ, π):

1. T_{s_1}: replace ℓ by another line ℓ' in π (sum over q choices).
2. T_{s_2}: replace π by another plane π' through ℓ' (sum over q choices).
3. T_{s_1}: replace ℓ' by another line ℓ'' in π' (sum over q choices).

Result: sum over all flags (ℓ'', π') reachable by a path $(\ell, \pi) \to (\ell', \pi) \to (\ell', \pi') \to (\ell'', \pi')$, where each step changes one component.

Right side: $T_{s_2}T_{s_1}T_{s_2}$. Starting from (ℓ, π):

1. T_{s_2}: replace π by π' through ℓ (q choices).
2. T_{s_1}: replace ℓ by ℓ' in π' (q choices).
3. T_{s_2}: replace π' by π'' through ℓ' (q choices).

Result: sum over all flags (ℓ', π'') reachable by a path $(\ell, \pi) \to (\ell, \pi') \to (\ell', \pi') \to (\ell', \pi'')$.

Both sides produce a sum over flags at "Bruhat distance 3" from (ℓ, π). The key point: a flag (ℓ'', π') is at distance 3 if and only if $\ell'' \neq \ell$ and $\pi' \neq \pi$ and $\ell'' \not\subset \pi$ and $\ell \not\subset \pi'$. This set—and the multiplicity with which each flag appears—is the same for both orderings, because the composition $s_1s_2s_1 = s_2s_1s_2$ in S_3 means the two paths through the Bruhat graph reach the same double coset $Bs_1s_2s_1B = Bw_0B$ (where w_0 is the longest element of S_3).

More precisely: both $T_{s_1}T_{s_2}T_{s_1}$ and $T_{s_2}T_{s_1}T_{s_2}$ equal T_{w_0} (the basis element for the longest Weyl group element), because in the Hecke algebra one has $T_w = T_{s_{i_1}} \cdots T_{s_{i_k}}$ whenever $w = s_{i_1} \cdots s_{i_k}$ is a reduced expression, and $w_0 = s_1s_2s_1 = s_2s_1s_2$ are the two reduced expressions. □

Remark 3.2. The proof relies on the fact that $\ell(s_1s_2s_1) = 3 = \ell(s_1) + \ell(s_2) + \ell(s_1)$, where $\ell(w)$ is the length of w (number of simple reflections in a reduced expression). This ensures that $T_{s_1}T_{s_2}T_{s_1}$ is a single basis element T_{w_0}, not a sum of basis elements. The multiplicativity $T_uT_v = T_{uv}$ holds whenever $\ell(uv) = \ell(u) + \ell(v)$.

4 The isomorphism

4.1 The Iwahori–Hecke algebra by generators and relations

Definition 4.1. The *Iwahori–Hecke algebra* $H_n(q)$ is the associative algebra over k generated by elements $T_1, \dots, T_{n-1}$ subject to:

$$(T_i - q)(T_i + 1) = 0 \qquad \text{for all } i, \tag{3}$$

$$T_i T_{i+1} T_i = T_{i+1} T_i T_{i+1} \qquad \text{for all } i, \tag{4}$$

$$T_i T_j = T_j T_i \qquad \text{for } |i - j| > 1. \tag{5}$$

At $q = 1$: the quadratic relation becomes $T_i^2 = 1$ and the algebra is $\mathbb{C}[S_n]$.

4.2 The isomorphism theorem

Theorem 4.2 (Iwahori, 1964)**.** *There is an algebra isomorphism*

$$H_n(q) \xrightarrow{\sim} \mathcal{H}(\mathrm{GL}_n(\mathbb{F}_q),\ B_n(\mathbb{F}_q)),$$

sending the generator T_i to the (suitably normalised) characteristic function of the double coset Bs_iB.

Proof sketch. We must verify that the double coset operators T_{s_i} satisfy the three relations of $H_n(q)$.

1. *Quadratic relation* (3): proved in Theorem 2.1 for $n = 2$. The same argument works for any i: the operator T_{s_i} acts on the i-th component of the flag (a line in a plane), and the counting on $\mathbb{P}^1(\mathbb{F}_q)$ gives $(T_{s_i} - q)(T_{s_i} + 1) = 0$.

2. *Braid relation* (4): proved in Theorem 3.1 for (s_1, s_2) in GL_3. The same argument works for any adjacent pair (s_i, s_{i+1}): both sides equal $T_{s_i s_{i+1} s_i}$ because the product is length-additive.

3. *Commutativity* (5): for $|i - j| > 1$, the operators T_{s_i} and T_{s_j} act on disjoint components of the flag (non-adjacent levels), so they commute.

It remains to show the map is an isomorphism. Both algebras have dimension $n!$: the Iwahori–Hecke algebra by its standard basis $\{T_w : w \in S_n\}$ (defined by $T_w = T_{s_{i_1}} \cdots T_{s_{i_k}}$ for any reduced expression), and the Hecke algebra by the Bruhat decomposition (Theorem 1.1). The map sends basis to basis. □

5 Summary

The Iwahori–Hecke algebra $H_n(q)$ has two faces:

Abstract (generators & relations)	**Geometric (double cosets)**
Generators $T_1, \ldots, T_{n-1}$	Double cosets Bs_iB
Quadratic $(T_i - q)(T_i + 1) = 0$	Counting on $\mathbb{P}^1(\mathbb{F}_q)$: q neighbours
Braid $T_iT_{i+1}T_i = T_{i+1}T_iT_{i+1}$	Flag geometry: two paths to w_0
Commuting $T_iT_j = T_jT_i$ for $\lvert i-j\rvert > 1$	Non-adjacent flag components
$q = 1$ gives $\mathbb{C}[S_n]$	$\mathbb{F}_1$: Borel = trivial, flags = permutations

The parameter q is the number of elements in $\mathbb{F}_q$, which is the number of points in a "local $\mathbb{P}^1$" (the lines in a plane, or the planes through a line). The quadratic relation $(T_s - q)(T_s + 1) = 0$ encodes this: T_s acting on $\mathbb{P}^1(\mathbb{F}_q)$ has eigenvalue q on the constant function and eigenvalue -1 on the complement. As $q \to 1$, $\mathbb{P}^1$ shrinks to two points, T_s becomes a transposition, and $H_n(q)$ degenerates to $\mathbb{C}[S_n]$.

References

[1] N. Bourbaki, *Lie Groups and Lie Algebras*, Chapters 4–6, Springer-Verlag, 2002.

[2] J. E. Humphreys, *Reflection Groups and Coxeter Groups*, Cambridge Studies in Advanced Mathematics 29, Cambridge University Press, 1990.

[3] N. Iwahori, *On the structure of a Hecke ring of a Chevalley group over a finite field*, J. Fac. Sci. Univ. Tokyo Sect. I **10** (1964), 215–236.